

SCIENTIFIC JOURNAL PUBLICATION TREND TRACKING SYSTEM

Group 6

I. INTRODUCTION

1 Overview

1.1 Project Information

- Project Name: Scientific Journal Publication Trend Tracking System
- Group Name: Group 6
- Software Type: Web Application
- GitHub Repository:
<https://github.com/NguyenNgocBaoTram10/Scientific-Journal-Publication-Trend-Tr>

2 Project Background

Describe the motivation and background of the project.

3 Scope

Describe the scope of the project.

4 Objectives

Describe the objectives of the project.

5 System Requirements

List the functional and non-functional requirements.

6 System Design

Describe the architecture, database design, and diagrams.

7 Implementation

Describe technologies, APIs, and key features implemented.

8 Testing

Describe test cases and testing results.

9 Conclusion

Summarize the project outcomes and future improvements.

9.1 Project team

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Bảng 1: Project Team Information

2 Functional Requirements

2.1 User Authentication and Authorization

- Users can register a new account.
- Users can log in using a valid username and password.
- The system assigns permissions based on user roles.
- Administrators can manage user accounts and roles.

2.2 Research Paper Search

- Users can search research papers by keyword, author name, or journal name.
- The system displays matching publication results.
- Users can view detailed publication information.

2.3 Bookmark Management

- Users can save research papers to personal bookmarks.
- The system prevents duplicate bookmarks.
- Users can access bookmarked papers later.

2.4 Publication Trend Analysis

- Users can analyze publication trends based on keywords, topics, authors, or journals.
- The system generates trend statistics from publication metadata.
- The system visualizes trends using charts and dashboards.

2.5 Dashboard Visualization

- Users can view publication statistics through dashboards.
- The dashboard displays publication counts, topic distributions, journal distributions, and growth trends.

2.6 Trending Topics

- Users can view trending research topics.
- The system ranks topics according to publication frequency and growth rate.

2.7 Topic Following

- Users can follow research topics of interest.
- The system stores followed topics for each user.
- Users may follow multiple topics.

2.8 Notification Management

- Users receive notifications related to followed topics.
- Notifications include new publications, trend changes, and emerging topics.
- Users can view notification details.

2.9 Report Generation

- Users can generate analytical reports based on publication trends.
- Reports summarize publication statistics and research trends.
- Reports are generated from publication metadata.

2.10 System Administration

- Administrators can manage system operations.
- Administrators can manage user accounts.
- Administrators can configure academic API sources.
- Administrators can synchronize publication metadata from external APIs.

2.11 Academic Data Synchronization

- The system synchronizes publication metadata from external academic APIs.
- The system validates retrieved metadata.
- The system inserts new records and updates existing records.
- The system records synchronization logs and statistics.

3 Non-Functional Requirements

3.1 Usability

The interface is intuitive and accessible for all user types.

3.2 Performance

Search results must be returned within 2 seconds.

The system must support at least 500 concurrent users.

3.3 Reliability

System uptime must be at least 99% during business hours.

Data must be backed up daily.

3.4 Security

User passwords must be hashed before storage.

Role-based access control must be enforced for all features.

3.5 Maintainability & Scalability

The system must use a modular architecture for easy maintenance.

New data sources can be added without redesigning the existing system.

4 Limitations of the Project

4.1 Limited Use of Full-Text Data

- The system only uses publication metadata such as title, abstract, keywords, authors, and publication year.
- It does not process full-text content, which limits the depth of analysis and understanding of research papers.

4.2 Restricted Data Scope

- The system focuses only on selected research fields (e.g., Computer Science and Artificial Intelligence).
- Therefore, it does not fully represent trends across all scientific disciplines.

4.3 Dependence on External APIs

- The system relies on external APIs such as Semantic Scholar, OpenAlex, and Cross-ref.
- Any API downtime, structural changes, or request limitations may affect system stability and data collection performance.

4.4 Simplified Trend Analysis and Update Frequency

- Trend analysis is mainly based on publication frequency and growth rate.
- It does not apply advanced Machine Learning or AI models for prediction.
- Data is synchronized periodically (daily or weekly), so real-time trends may not be immediately reflected.

II. PROJECT MANAGEMENT PLAN

1 Overview

1.1 Scope and estimation

#	WBS Item	Complexity	Duration (Days)
1. Project Initiation			
1.1	Define project scope	Medium	1
1.2	Collect and analyze requirements	Complex	2
2. Project Management and Environment Setup			
2.1	Set up development environment (GitHub, Database, Backend, Frontend)	Simple	1
2.2	Design database (ERD, Database Schema)	Complex	2
2.3	Configure security (JWT, RBAC, Password Hashing)	Complex	2
3. System Design			
3.1	Design Use Case Diagram and Activity Diagram	Medium	1
3.2	Design Sequence Diagram	Medium	1
3.3	Design DFD and ERD	Medium	1
4. System Development			
4.1	User Registration and Login	Medium	2
4.2	Search Scientific Publications	Complex	2
4.3	View Publication Details	Medium	1
4.4	Bookmark Management	Medium	1
4.5	Research Topic Following	Medium	1
4.6	Notification Management	Medium	1
4.7	Publication Trend Analysis	Complex	3
4.8	Develop Statistical Dashboard	Complex	2
4.9	Display Trending Topics	Medium	1
4.10	Generate Statistical Reports (PDF/CSV)	Complex	2

#	WBS Item	Complexity	Duration (Days)
5. Administration Functions			
5.1	User Management	Complex	2
5.2	Configure Academic API Sources	Complex	2
5.3	Synchronize Data from Semantic Scholar, OpenAlex, and Crossref	Complex	3
6. Testing			
6.1	Prepare Test Plan	Medium	1
6.2	Design Test Cases	Medium	2
6.3	Execute Testing	Complex	2
6.4	Fix Defects After Testing	Complex	2
7. Project Finalization			
7.1	Complete Project Documentation	Medium	2
7.2	Deploy the System	Medium	1
7.3	Prepare Presentation Slides	Simple	1
Total Duration			46 Working Days